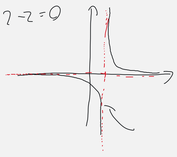
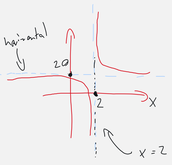
$h(x) = \frac{1}{x-2}$, $x \neq 2$

VA : $x = 2$
HA : $y = 0$

$x = 2 \Rightarrow 2 - 2 = 0$

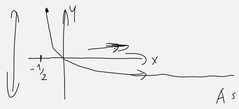




$f(x) = \frac{3}{\sqrt{2x+1}} - 2$, $x > -\frac{1}{2}$

$\Rightarrow -0.5$

-0.49

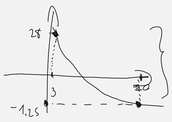


As $x \Rightarrow -\frac{1}{2} \Rightarrow y \Rightarrow \infty$

As $x \Rightarrow \infty \Rightarrow y \Rightarrow -2$

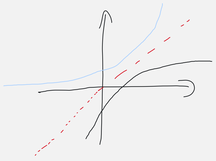
$\frac{3}{\sqrt{2x+1}}$ → ∞ , -2

Range : $(-2, \infty)$



Range : $[-1.25, 28]$

Range : $[2.73, 10]$



$y = 2x + 5$

$x = 2y + 5$

$\frac{x-5}{2} = y$

	Domain	Range
$f(x)$	$(2, 10)$	$(0, \infty)$
$f^{-1}(x)$	$(0, \infty)$	$(2, 10)$

$3 \leq y \leq 20$

$3 \leq f^{-1}(x) \leq 20$

$f(x) = \frac{300}{x^2+1} - 2$, $3 \leq x \leq 20$,

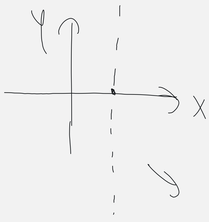
$-1.25 \leq y \leq 20$

(a) domain $f^{-1}(x)$ → $-1.25 \leq x \leq 20$

Write dom

domain $f^{-1}(x)$ = range $f(x)$

range domain



↘ $x = 1$

$13 = 3 + \frac{15}{3x+3}$

↘ $f(x) = 3 + \frac{15}{3x-3}$

$f(13)$ $f^{-1}(13)$

↘ $13 = 3 + \frac{15}{3x-3}$

↘ $3 + \frac{15}{3 \cdot 13 - 3}$

↘ x

$x^2 = 49$

$x = \pm 7$ ↘ $7^2 = 49$

$x = 7$ $(-7)^2 = 49$

$a^2 = 4$

$a = \pm 2$

$2^2 = 4$

$(-2)^2 = 4$

(b) $f^{-1}(4) = 7$

↘ 4 length

(c)

10kg mass cannot → 50kg lag

$C(10) = 50$

↘ x ↘ C

$C^{-1}(4) = 7$

↘ x

↘ C

↘